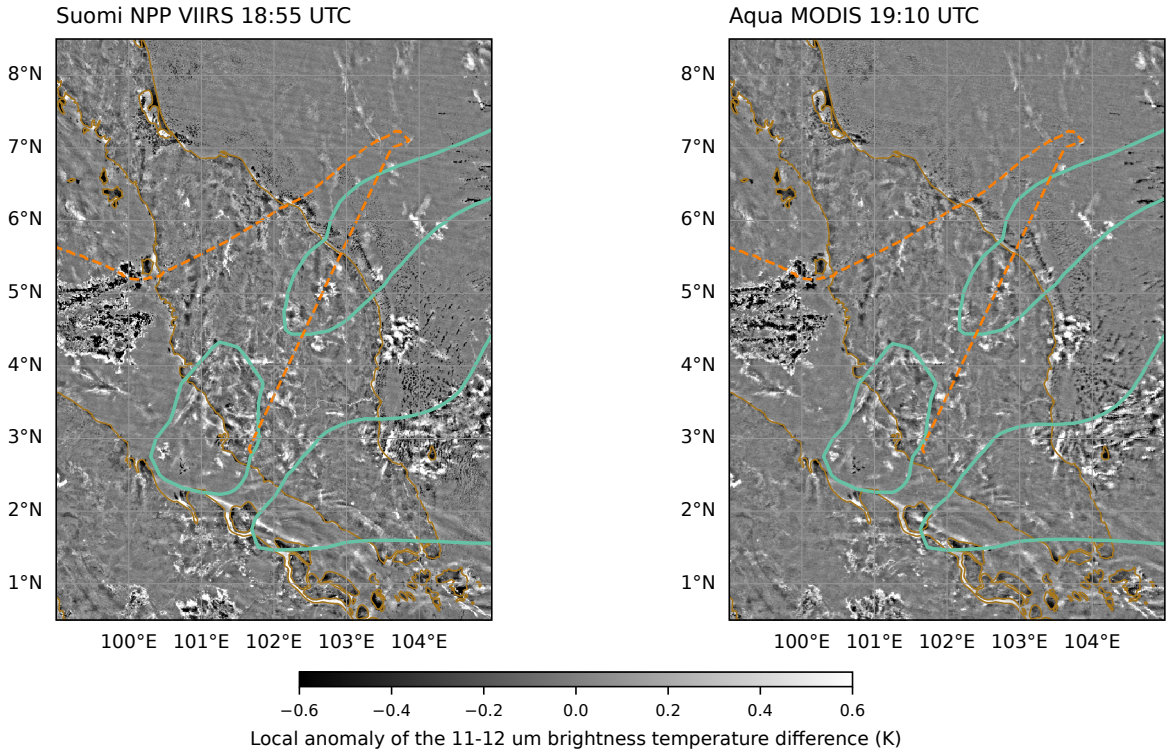
No contrail or thin cirrus visible in the ice-supersaturated layer at FL400



edge of ice-supersaturated air at FL400 (ERA5, scan hour)

radar-tracked flight path, 16:42-18:22 UTC

Plain note: the green line encloses air that was moist enough at 40,000 ft for a contrail to last. Inside it the pictures show no contrail and no thin ice cloud, so no aircraft trail is available as a test of detectability.

Technical note: images as in the pass figures (VIIRS M15-M16, VNP02MOD/VNP03MOD A2014066.1854.002; MODIS 31-32, MYD021KM/MYD03 A2014066.1905/.1910.061), split-window difference minus a 21-pixel running median, 0.01 deg grid. Ice-supersaturation: ERA5 (ARCO mirror) T and q at 187.5 hPa interpolated in log-p, linearly in time to 18:55 and 19:10 UTC; RHi from Murphy and Koop (2005); contour RHi = 1. Supersaturated area at 187.5 hPa = 8.6 % of the 91.5-105 E, 0-9.5 N scene at 18:55 (175 hPa 6.5 %, 200 hPa 4.8 %, 225 hPa 0 %). No cold cloud (11 um brightness temperature < 235 K) inside the supersaturated area in either image (fraction 0.000). Bright texture over land is judged to be land-surface emissivity contrast, not cloud. Radar track from the Geoscience Australia digital military radar data. ERA5 humidity at these levels has known biases; position of the contour is uncertain by tens of km.